Multi-Omics and Multi-Modal Fusion-Based Reporting System for Flavor Fingerprint Evolution and Intelligent Processing Decisions in Condiments

1. Project Overview

Traditional Chinese condiments (vinegar, soy sauce, sauces, etc.) carry deep culinary-cultural value and a substantial industrial base, with the industry's total scale approaching RMB 600 billion (China Condiment Association, 2024 Top 100 Renowned Condiment Brand Enterprises Data). However, under open multi-strain solid-state fermentation systems, flavor formation in these products is jointly regulated by multiple coupled factors such as raw materials, microbial succession, processing conditions, and aging environment (Li et al., 2025, Food Research International; Xia et al., 2022, Frontiers in Microbiology; Wu et al., 2024, Food Science and Human Wellness), and long-standing issues persist with poor batch-to-batch stability and quality evaluation that depends on veteran masters' experience-based judgment — “look, smell, taste, touch” (Li et al., 2025, Food Research International; Jo et al., 2016, Food Science and Biotechnology). As consumer demand for consistent quality and differentiated flavor grows, the traditional condiment industry is entering a critical window for transformation from “experience-based manufacturing” to “digital intelligent manufacturing.”

Taking traditional vinegar as the entry point, this project is grounded in the cross-disciplinary direction of “food flavoromics + artificial intelligence + intelligent processing quality control,” aiming to build an AI-based flavor fingerprint evaluation and intelligent quality-control system for traditional condiments. The project seeks to integrate physicochemical indicators, volatile/non-volatile flavor compounds, electronic-nose/electronic-tongue sensor data, microbial community information, and human sensory evaluation results (Li et al., 2022, Metabolites; Zhou et al., 2020, Food Research International; Liang et al., 2016, Journal of Agricultural and Food Chemistry; Zhou et al., 2017, Journal of Chromatography A; Zhu et al., 2016, Journal of Food Science and Technology) to establish a multi-dimensional digital flavor indicator system, converting traditional sensory descriptors such as “aroma, acidity, mellowness, body, and agedness” into measurable, modelable, and interpretable digital indicators.

On the technical roadmap, the project builds a four-layer closed-loop architecture: “data collection → omics analysis → flavor prediction → Agent-automated report generation.” First, a baseline flavor database is built from representative vinegar samples; second, methods such as OAV and TAV are combined to weight flavor compounds by their sensory contribution (Zhou et al., 2020, Food Research International; Liang et al., 2016, Journal of Agricultural and Food Chemistry); then, multiple machine-learning methods are used to build models for flavor-quality classification, aging-maturity prediction, and abnormal-batch identification, combined with interpretability techniques such as SHAP to trace back key flavor-contributing factors (Wang et al., 2026, npj Science of Food; Anwar et al., 2023, review on food quality assessment; Arrighi et al., 2025, explainable AI for food datasets). The most critical improvement lies in encapsulating the above expert workflow — data-cleaning conventions, feature-selection logic, model-invocation procedures, result-interpretation rules, and exception-handling experience — into standardized Skill interfaces that a large language model (LLM) Agent can automatically orchestrate and execute: based on new batches of upstream metabolomic data, the Agent can automatically call the corresponding Skills in series or in parallel, completing the end-to-end automated generation of a flavor-evaluation report from raw mass-spectrometry data to final output — including a flavor radar chart, ranked contributions of key metabolites, abnormal-sample alerts, and confidence intervals in a standardized quality-control report. This greatly improves the efficiency, reliability, and stability of data analysis.

The core innovations of this project are reflected in four aspects: (1) Methodological innovation — converting traditional experience-based judgment into a quantifiable, reusable data asset, rather than simply replacing human labor; (2) Data-fusion innovation — building a multi-modal flavor feature system that integrates metabolomics, sensor data, and sensory evaluation, overcoming the limitations of any single detection method; (3) Interpretability innovation — introducing explainable-AI methods such as SHAP so that the model's decision-making process is transparent and understandable to enterprise users without a data-science background; (4) Engineering innovation (the core innovation) — for the first time encapsulating the expert knowledge system for flavor quality control of traditional condiments into a Skill library callable by AI Agents, turning analysis workflows that once depended on manual, step-by-step operation into an automated, reusable process, substantially improving the efficiency and consistency of flavor-report generation. The Skill library is also extensible:

individual Skills can be iterated based on expert feedback or new data characteristics without rebuilding the entire pipeline, truly achieving a closed intelligent loop in which "expert experience can be captured, analysis workflows can be reused, and quality-control decisions can be traced."

The project carries clear scientific value (advancing traditional flavor evaluation from experience-based to standardized), industrial value (providing enterprises with an objective quality-control tool that reduces batch-variation risk), and social-cultural value (using digital means to preserve traditional fermentation craftsmanship and helping heritage brands transition toward new productive forces).

In summary, taking traditional vinegar as the breakthrough point, this project builds a complete closed-loop solution — from "flavor data collection → flavor fingerprint construction → AI model evaluation → interpretable analysis → Agent-automated report generation" — providing the traditional condiment industry with a deployable, scalable, and transferable intelligent technical pathway for upgrading digital quality control.

2. Main Research Content and Technical Innovations

2.1. Agent

This project builds an automated analysis and reporting system centered on Agent technology, using unified scheduling to link four stages — data mining, model evaluation, report generation, and knowledge innovation — into an automated closed loop of "data input → intelligent analysis → report output → new-knowledge feedback." The system consists of the following four collaborative subsystems:

2.1.1. Flavor Fermentation Omics Data Analysis Subsystem (Data Middle-Platform Layer)

Responsible for underlying data governance and feature extraction. It connects to raw spectra output by instruments such as GC-MS/LC-MS and automatically performs baseline correction, peak alignment, compound identification, and relative quantification. Within it, a Data-Cleaning Agent monitors newly ingested data and triggers the standardized processing pipeline, while a Feature-Engineering Agent uses the expert knowledge base to screen differential metabolites strongly associated with flavor attributes, outputting a structured "compound–abundance" matrix for downstream use.

2.1.2. Machine-Learning Evaluation and Invocation Subsystem (Decision-

Analysis Layer)

Responsible for dynamically selecting and invoking the optimal evaluation model. It has a built-in library of multivariate statistics, cluster analysis, and various flavor-scoring models; a Scheduling Agent automatically recommends a suitable machine-learning pipeline based on data volume, distribution characteristics, and analysis goals. A Model-Selection Agent uses meta-learning to quickly estimate each model's expected performance, and a Hyperparameter-Optimization Agent performs automatic tuning, ultimately producing a cross-validated model-performance report and a variable-importance ranking.

2.1.3. Report-Rendering Subsystem (Human–Machine Interaction Layer)

Responsible for turning analysis results into professional deliverables for decision-makers. It supports multiple output formats, including interactive HTML dashboards and PDF research reports, with content covering clustering heatmaps, VIP-compound rankings, and trend visualizations. A Report-Orchestration Agent automatically matches layout logic to preset templates, while a Narrative Agent translates data fluctuations into business language. The subsystem also supports a human–machine collaborative review mechanism, feeding expert revision records back into the template library for continuous improvement.

2.1.4. New-Method Discovery Subsystem (Knowledge-Evolution Layer)

Responsible for the system's continuous evolution. Through deep mining of historical tasks, model performance, and expert revision records, it automatically discovers better feature combinations, candidate novel flavor markers, and new analysis methods, encapsulating them as new Skills for other subsystems to call — enabling autonomous learning and evolution.

The four subsystems are linked by a central Scheduling Agent in an event-driven manner: data ingestion → triggers preprocessing → automatically invokes model evaluation → evaluation results feed report generation and knowledge mining in parallel → the report is delivered while method suggestions are written to the knowledge base. The entire process requires no manual intervention, with expert confirmation introduced only at high-confidence nodes, achieving automated transformation from raw data to a decision-ready report.

2.2. Flavor Fermentation Products

The flavor fermentation products referred to in this project are open, multi-strain, solid-state-fermented condiments represented by vinegar. Vinegar was chosen as the entry point partly because its flavor system is typical yet complex — with a full range of sensory characteristics such as aroma, acidity, mellowness, body, and agedness — making it an ideal vehicle for studying "how flavor is characterized at the molecular level"; and partly because vinegar has relatively rich public metabolomic and microbiome data and clearly defined fermentation and aging stages, which makes it easy to build a reproducible analysis pipeline that can later be transferred to similar fermented condiments such as soy sauce, rice wine, and broad bean paste.

In terms of composition, the flavor of vinegar results from the superposition of volatile and non-volatile components. Volatile components determine the aroma one "smells," mainly small molecules such as esters, aldehydes, ketones, pyrazines, furans, and volatile acids; non-volatile components determine the taste one "tastes," mainly organic acids, amino acids, sugars and sugar alcohols, peptides, phenolics, and biogenic amines. Together these hundreds to thousands of substances form a "metabolic fingerprint," which is also why characterizing flavor fermentation products inherently requires the complementary use of both GC-MS and LC-MS platforms rather than relying on a single method.

In terms of formation mechanism, the flavor of fermentation products is not static but is dynamically generated under the coupled effects of raw-material composition, microbial community succession, processing conditions, and aging environment. Across the stages from saccharification, alcoholic fermentation, and acetic fermentation to post-maturation aging, microbial metabolism together with Maillard and esterification reactions continuously changes the types and levels of flavor compounds: for example, fresh vinegar is relatively high in fruity esters, which gradually volatilize and are lost during aging, while Maillard products such as tetramethylpyrazine rise significantly, giving aged vinegar a more mellow, complex aroma.

It is precisely this “process—microorganism—chemistry—flavor” causal chain that provides the scientific basis for converting sensory descriptions into modelable rules of molecular evolution.

From a modeling perspective, flavor fermentation products have a pronounced time dimension and batch-specific character. Fermentation stage, aging duration, and production season all systematically alter the metabolic fingerprint, providing a natural timeline for modeling “flavor fingerprint evolution”; at the same time, the batch variation introduced by open solid-state fermentation makes flavor stability an industry pain point. This project takes such flavor fermentation products as its research object, using metabolomics and flavoromics to obtain their molecular-level fingerprints, which are then passed to downstream machine learning and Agents to complete flavor evaluation and process decision-making — upgrading flavor control that once “relied on veteran masters' experience” into an objective, quantifiable, and traceable digital capability.

2.3. Metabolomics and Flavoromics

The flavor of vinegar and other flavor fermentation products is jointly determined by hundreds to thousands of volatile and non-volatile small molecules, which form the material basis for grading, blending, and aging judgments; yet for a long time the composition and evolution of these flavor compounds has relied mainly on masters' sensory experience, lacking an objective, quantifiable molecular-level characterization, and the quantitative relationship between process parameters and flavor has remained unclear. Metabolomics can measure the relative content of these small molecules in a single assay, yielding an objective “metabolic fingerprint” that provides a material-level entry point for the digital characterization of flavor.

To address the above pain points, this section uses metabolomics to obtain the metabolic fingerprint of flavor fermentation products and, with the help of sensory thresholds, upgrades it into a flavoromics interpretation. The core research focuses on three levels:

① a panoramic characterization of flavor compounds using complementary dual mass-spectrometry platforms, with GC-MS characterizing aroma-related volatiles such as esters, aldehydes, ketones, pyrazines, and volatile acids, and LC-MS characterizing taste-related non-volatiles such as organic acids, amino acids, sugars, and phenolics; ② a reproducible analysis chain running from raw spectra to differential metabolites and then to pathway mechanisms, answering questions such as "can different vinegars be objectively distinguished, which compounds drive the distinction, and which metabolic pathways the differences fall into"; ③ weighting differential compounds using sensory thresholds such as odor activity value and taste activity value, translating statistically differential compounds into the key compounds that truly affect flavor and attributing them to mechanisms such as Maillard reaction, esterification, and fermentation metabolism.

In terms of methodological innovation, this section builds a standardized upstream pipeline that "chooses software by instrument" (Compound Discoverer for Thermo data; the open-source, cross-platform MS-DIAL for Agilent/AB Sciex data, etc., with an independent peak-extraction script built on pymzML as a reproduction path that does not depend on commercial software), paired with log2-transform-plus-Pareto-scaling preprocessing, PCA and OPLS-DA (VIP) discrimination, differential screening via the Kruskal-Wallis test with BH-FDR correction, and an offline KEGG pathway-enrichment downstream chain; it further performs partial least-squares regression of aging duration against the OAV profile of key flavor compounds, building a "flavor clock" quantitative model that infers aging year from the flavor fingerprint. The entire analysis pipeline is consolidated into parameter-traceable, reusable Skills for stable orchestration and invocation by the upper-level Agent.

Preliminary validation on public real-world data shows that five types of grain vinegar can be clearly distinguished by GC-MS fingerprint alone; the "flavor clock" can infer the aging duration of Zhenjiang aromatic vinegar with an error of about half a year; and differential-metabolite enrichment points to pathways such as phenylalanine/tyrosine metabolism, explaining the flavor differences among vinegar types at the mechanistic level. These results convert vague sensory descriptors such as “aroma, acidity, mellowness, body, and agedness” into detectable, modelable, and interpretable molecular indicators — serving both as the metabolite layer of the multi-omics flavor fingerprint and as the material basis and feature input for downstream machine-learning flavor prediction and intelligent quality control.

2.4. Machine Learning

In the industrial production of flavor fermentation products, “tasting by veteran experience” remains the dominant approach to sensory quality control, with long-standing pain points including strong subjectivity, poor reproducibility, high labor cost, and an inability to quantitatively guide process adjustment. At the same time, enterprises have already accumulated large volumes of GC-MS/LC-MS omics data and electronic-tongue response spectra, which remain “dormant data” due to high dimensionality and complex nonlinear interaction effects, and have not been effectively converted into quality-control decision value.

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To address this contradiction, this project uses human expert taste scores as the supervision target and carries out machine-learning modeling research on flavoromics and electronic-tongue multi-modal data. The core research focuses on three levels: ① mining stable mapping relationships between high-dimensional, sparse omics data and sensory labels; ② nonlinear characterization of how synergistic/antagonistic effects among compounds contribute to overall flavor; ④ ensuring the chemical interpretability of model decisions, so that outputs are consistent with food-science common sense and meet compliance-review requirements.

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In terms of methodological innovation, this project moves beyond the expressive bottleneck of traditional linear methods such as partial least-squares regression (PLSR), building a multi-modal fusion deep-learning framework that combines a molecular graph neural network (GNN) with a spectral Transformer, achieving end-to-end representation learning from molecular structure to the overall sensory score of a mixture. To address the practical constraint of limited sample sizes in the flavor domain, it introduces physics-informed regularization and a contrastive-learning pretraining strategy to suppress overfitting and

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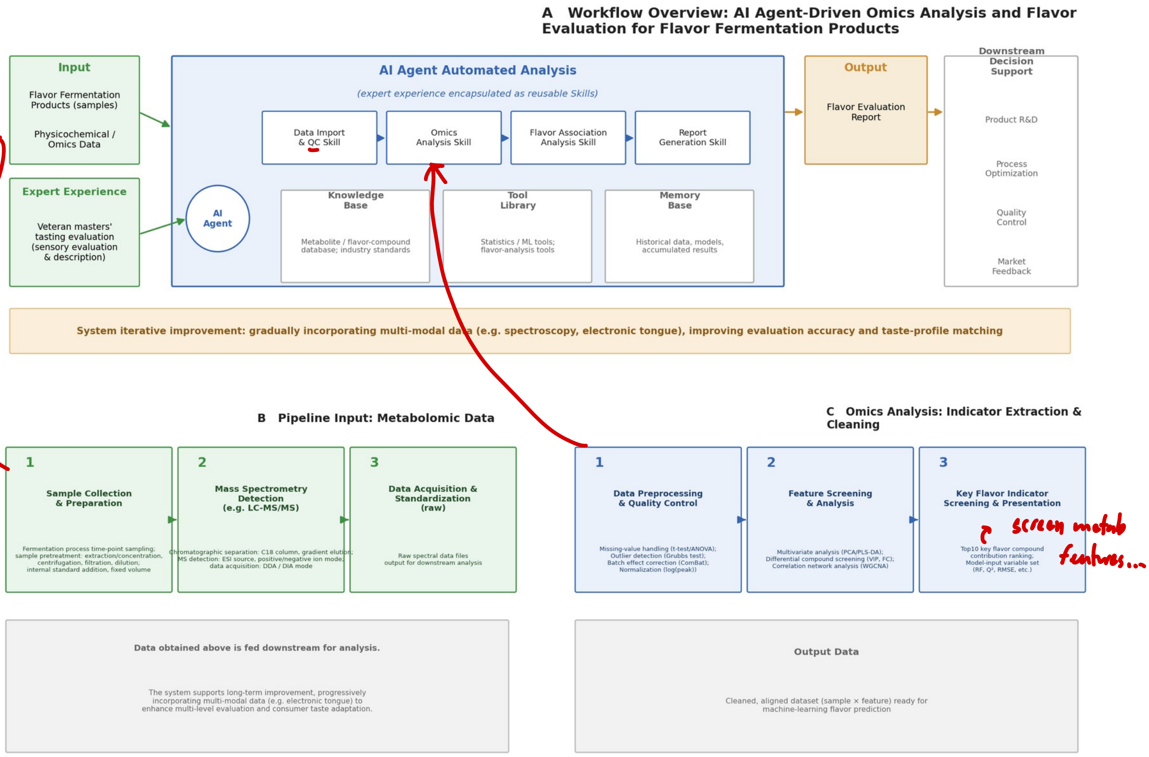
improve generalization under limited labeled data; it also integrates Grad-CAM and SHAP attribution modules, giving the deep model's predictions atom-level and compound-level interpretability, filling the gap of “untrustworthy black boxes” that has limited deep learning in flavor sensory prediction.

Ultimately, this modeling approach is expected to raise the prediction accuracy of key sensory dimensions (acidity, sweetness, bitterness, fruity aroma, etc.) from the traditional methods' $R^2 \approx 0.60–0.70$ cite? to above 0.85, ?? effectively supporting practical industrial scenarios such as intelligent determination of the fermentation endpoint, flavor-fluctuation warnings after raw-material substitution, and recommendations for formula fine-tuning — upgrading the experience-driven “post-hoc inspection” into data-driven “real-time prediction,” and providing core technical support for the transition of flavor quality control from experience-dependent to data-intelligent.

3. Technical Route

This project is based on upstream metabolomic data of flavor fermentation products. By encapsulating human experts' workflows and professional experience into reusable Skills, the analysis is automatically executed by an AI Agent, ultimately generating a flavor-evaluation report that provides a reference basis for downstream decision-making (Figure 1A).



We divide the workflow into the following modules for research, development, and translation; this section mainly introduces the workflow, while the implementation pathway is covered in later subsections.

3.1. Pipeline Input

The pipeline's main input is the metabolomic data of flavor fermentation products, i.e., raw mass-spectrometry spectra (Figure 1B). Since vinegar flavor spans both volatile aroma and non-volatile taste, data are sourced from two complementary platforms: gas chromatography–mass spectrometry (GC-MS, with headspace solid-phase microextraction injection) for aroma-related volatiles such as esters, aldehydes, ketones, pyrazines, and volatile acids, and liquid chromatography–mass spectrometry (LC-MS, non-targeted acquisition) for taste-related non-volatiles such as organic acids, amino acids, sugars and sugar alcohols, and phenolics, with positive- and negative-ion modes injected separately to jointly form a panoramic input of flavor compounds; quantitative data from the literature or targeted assays are added as supplements where necessary. The data obtained above are fed downstream for analysis after unified naming and format verification.

Besides the inputs mentioned above, the system also has the capacity for long-term improvement, and can progressively incorporate multi-modal data such as electronic-tongue inputs, giving the system multi-level capability to evaluate fermentation products, thereby continuously increasing the system's evaluation accuracy and its adaptation to consumers' multi-level taste preferences over the long term.

3.2. Omics Data Analysis

Using the data measured above, the system processes and analyzes omics data based on statistical and machine-learning methods to extract effective indicators and clean the data (Figure 1C). Because the metabolome of flavor fermentation products is characterized by high feature dimensionality, coexisting noise and missing values, and large within-group variation, this stage proceeds through the sequence "upstream feature extraction → data cleaning and preprocessing → multivariate statistics and differential analysis → pathway enrichment and mechanistic interpretation → flavoromics key-compound weighting → effective-indicator output and Skill encapsulation," with every sub-step implemented in a standardized, parameter-traceable way to ensure stable, reproducible results.

3.2.1. Upstream Feature Extraction: From Raw Spectra to the Metabolite Matrix

The upstream stage is responsible for converting raw spectra output by vendor instruments into a feature matrix suitable for statistics, following the principle of "choosing software by instrument": Compound Discoverer is preferred for Thermo data, while the open-source, cross-platform MS-DIAL is used for Agilent, AB Sciex, Waters, and similar instruments; vendor binary formats such as .raw/.wiff/.d are first uniformly converted to an open format such as mzML, ensuring that data from different instrument sources can be read by the same pipeline. After import, the system suppresses noise in each spectrum by signal-to-noise ratio and identifies genuine m/z signals, then links the same m/z along retention time into chromatographic peaks, algorithmically splits co-eluting peaks, performs retention-time correction and missing-value imputation across multiple samples, and merges the positive- and negative-ion mode library searches to obtain a complete feature matrix.

On this basis, the system annotates metabolites using databases such as HMDB, KEGG, and the GC-MS NIST library together with retention index (RI), and accurately distinguishes between the “number of detected features” and the “number successfully annotated.” To demonstrate that the pipeline is not locked to commercial software and that results are robust and transferable, in addition to the alignment results from the above software, the system also independently reproduces the upstream stage from raw mzML files using a custom peak-extraction script built on Python's pymzML. Both paths ultimately produce a structured “sample × metabolite” matrix for downstream analysis.

3.2.2. Data Cleaning and Preprocessing

To eliminate instrument drift and scale differences and improve the robustness of subsequent statistics, the system first cleans the feature matrix: samples with obvious experimental or instrument failure are removed outright, and remaining missing values are imputed with the median or minimum for small samples, or predicted via methods such as KNN or random forest for large samples. Normalization and data transformation follow: TIC or median normalization is used for relative quantification, log2(x+1) stabilizes variance, and Pareto scaling balances high- and low-abundance compounds; this pipeline uses the log2-plus-Pareto combination by default.

Throughout preprocessing, equal-volume pooled QC samples are used to monitor reproducibility and instrument drift, and unstable features are filtered using a relative standard deviation (RSD/CV) threshold below 30%; for batch effects, PCA is first used to diagnose their presence, and effects are controlled through experimental design choices such as unified storage and single-batch injection, rather than forcibly merging different batches, to avoid introducing spurious differences.

3.2.3. Multivariate Statistics and Differential Analysis

On the cleaned matrix, the system combines unsupervised and supervised strategies to pinpoint the key metabolites that distinguish samples. PCA is used first for unsupervised exploration to observe overall distribution, batch structure, and outliers, at which point QC samples should cluster tightly together; because within-group variation is large, PCA often cannot separate groups — a common situation in metabolomics that signals the need for supervised methods. PLS-DA, or OPLS-DA (which is better suited to metabolomics), is then used for group discrimination, computing variable importance in projection (VIP) and reporting R^2 together with Q^2 obtained via leave-one-out cross-validation to assess model fit and predictive power, supplemented where necessary by permutation testing to guard against overfitting.

Beyond multivariate discrimination, the system simultaneously performs univariate differential analysis: t-tests for two-group comparisons and ANOVA for multi-group comparisons, switching to the more robust Kruskal-Wallis or Wilcoxon test when normality is not met, paired with Fold Change to gauge magnitude of change and Benjamini-Hochberg correction for multiple testing. The final set of differential metabolites is 2) determined by the priority 1) "statistical significance first, then magnitude of change, and a VIP constraint added if there are too many differences," using the combined criteria of $p < 0.05$, Fold Change above a threshold, and VIP > 1.

3.2.4. Pathway Enrichment and Visualization

To elevate differential metabolites from a mere "list of compounds" to "pathway mechanisms," the system performs Fisher's exact test-based over-representation analysis with FDR correction on the differential compounds that have KEGG IDs, using all annotated metabolites detected in the experiment as the background set, and transparently reports the choice of background set in the methodology, since a different background set directly affects enrichment conclusions. Results are presented as standardized charts, including PCA and PLS-DA score plots, VIP bar charts, volcano plots drawn from all metabolites, differential-compound clustering heatmaps, and pathway-enrichment bubble plots, forming graphical assets that can go directly into the report.

3.2.5. Flavoromics Interpretation and Key-Compound Weighting

Statistically differential metabolites are still one step removed from "flavor" and need to be translated, with the help of sensory thresholds, into the key compounds that truly affect flavor. The system computes the odor activity value (OAV, concentration divided by the olfactory threshold) for aroma compounds and the taste activity value (TAV, content divided by the taste threshold) for taste compounds; only compounds with an activity value greater than 1 are counted as actually contributing to flavor, screening out the true key flavor compounds from among the many differential compounds.

Building on this, the system classifies key compounds by chemical category — esters, acids, aldehydes, ketones, pyrazines, furans, etc. — mapping them to mechanisms such as Maillard reaction, esterification, and fermentation metabolism, clarifying the causal chain from process to chemistry to flavor; it also performs partial least-squares (PLS) regression of aging duration or fermentation stage against the key-flavor-compound matrix, building a "flavor clock" model that infers process indicators from the flavor fingerprint, providing downstream machine learning with flavor features that carry clear business meaning.

3.2.6. Effective-Indicator Output and Skill Encapsulation

The stages above ultimately produce effective indicators that downstream machine learning can call directly, including the "sample × metabolite" abundance matrix, differential metabolites with their VIP, Fold Change, and p-values, a list of significant pathways, and activity-value features of key flavor compounds. To ensure stable, reusable workflows, the system encapsulates every analysis stage as a Skill with a single responsibility, clearly defined input/output specifications, and traceable parameters, orchestrated and invoked in series or in parallel by the upper-level Agent as needed — achieving automated, reproducible computation from raw spectra to effective flavor indicators, and laying a quantitative foundation for expert review and subsequent iteration.

3.3. Machine-Learning Flavor Prediction

3.3.1. Baseline Models: Regularized Multivariate Statistics and Ensemble Learning

As a performance benchmark, traditional machine-learning baselines are established first:

Sparse linear models: Elastic-Net and Lasso regression are used, applying L1/L2 mixed regularization to screen key flavor markers under high-dimensional data, producing a variable-importance ranking with clear chemical meaning.

Nonlinear tree models: LightGBM and random forest are introduced to capture interaction effects among compounds through ensemble learning; their built-in feature-importance estimates provide a structured prior for subsequent deep models.

The baseline models serve two purposes: ① verifying data quality and label consistency; ② providing a reference threshold for the performance gain of deep models (if deep learning does not significantly outperform LightGBM, the approach falls back to a tree-model solution).

3.3.2. Core Model: Deep Representation Learning for Flavor Perception

To move beyond the expressive bottleneck of linear models for flavor interaction effects, a dedicated deep neural network architecture for sensory prediction is built:

Molecular fingerprint encoder (molecular level): converts identified compound structures into ECFP4 (extended-connectivity fingerprints) or SMILES sequences, and uses a graph neural network (GNN) to perform message passing over the molecular graph structure, learning the microscopic contribution of atom-level features to sensory attributes.

Omics-spectrum encoder (mixture level): for multi-dimensional spectral vectors output by GC-MS/electronic tongue, a 1D-CNN plus multi-head self-attention (Transformer Encoder) architecture is used to automatically extract local correlation patterns and long-range dependencies between adjacent retention times or sensor channels.

Cross-modal fusion layer: molecular-graph features and spectral features are adaptively fused via a cross-modal attention mechanism, allowing the model to attend simultaneously to “which compounds are present” and “the overall distribution shape of these compounds in the spectrum,” before a fully connected layer maps the result to specific sensory-dimension scores (e.g., bitterness, fruity-aroma intensity, mellowness).

3.3.3. Regularization and Prior Constraints for Small Samples

To address the pain point of limited sample sizes in flavoromics data (typically ≤200 batches), the following strategies are introduced to suppress overfitting:

Physics-informed regularization: known flavor synergy/antagonism relationships (e.g., the sugar-to-acid ratio threshold range) are encoded as penalty terms, constraining network weights to optimize in directions consistent with food-chemistry common sense.

Contrastive-learning pretraining: unlabeled electronic-tongue/mass-spectrometry data are used to build self-supervised pretraining tasks (e.g., masked spectrum reconstruction), so the encoder first learns a general representation of spectra before fine-tuning on a small labeled dataset, enabling few-shot learning.

Monte Carlo Dropout: Dropout is enabled at inference time, performing multiple random forward passes on the same input; the variance of the resulting prediction distribution is used to assess the model's confidence for a given sample, serving as a trigger signal for subsequent human–machine collaborative review.

3.3.4. Interpretability Attribution Module (XAI)

To dispel “black box” concerns and meet food-industry compliance requirements, the model must be interpretable:

Gradient-weighted Class Activation Mapping (Grad-CAM): applied to the 1D-CNN layer to locate the retention-time window or electronic-tongue sensor channel that contributes most to the final score, producing a visual card of “heatmap overlaid on spectrum.”

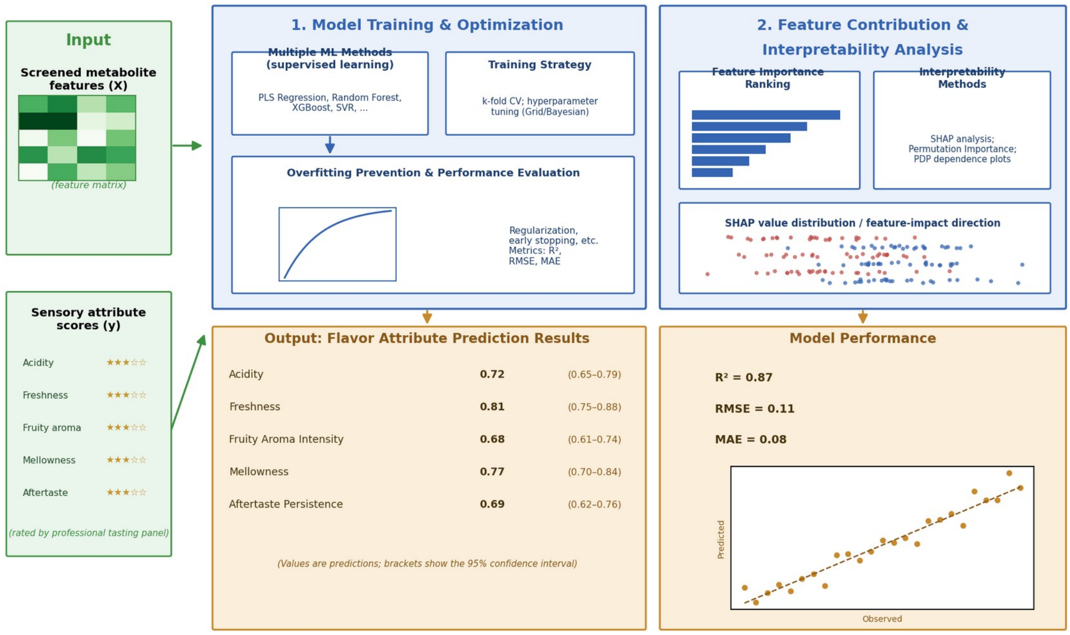
SHAP value back-propagation: SHAP (Shapley Additive Explanations) values from the deep model on the test set are mapped back to the original compounds, forming a contribution ranking of “key flavor compound → sensory score” and ensuring that the markers uncovered by the deep model are consistent with established knowledge or the literature.

Counterfactual explanation: answers questions such as “which compounds' abundances would need to change to raise the fruity-aroma score from 6.5 to 8.0?” — providing direct data guidance for process optimization.

3.3.5. Model Evaluation and Selection Criteria

Dimension	Evaluation Metric	Selection Principle
Prediction accuracy	RMSE, R^2, MAE	LightGBM sets the bar; a deep-learning model is adopted only if it improves R^2 by ≥ 0.08 on the validation set
Generalization stability	Leave-one-batch-out cross-validation	Different samples from the same production batch must be strictly separated to prevent data leakage from producing spuriously high accuracy
Inference efficiency	Per-sample inference time	Kept at ≤100 ms to meet real-time batch-release decision needs

After completing the preprocessing of omics data and extraction of effective indicators, the system further builds a flavor-prediction model, establishing a quantitative association between the metabolite profile and sensory attributes. Specifically, the screened metabolite features are used as input variables, and flavor-attribute scores (such as acidity, freshness, fruity aroma intensity, mellowness, etc.) calibrated by a professional tasting panel are used as prediction targets; multiple machine-learning algorithms are used for supervised modeling (Figure 2).



Tracing decisive metabolites via feature-contribution values provides interpretable support for sensory evaluation, and also establishes a quantitative foundation for expert review and Skill encapsulation.

During model training, the system uses cross-validation, feature-importance ranking, and hyperparameter tuning to improve generalization while preventing overfitting. The final output includes predicted values for key flavor indicators along with their confidence intervals.

Based on feature contributions from the training process, interpretability methods can also be used to trace back which metabolites play a decisive role in specific flavor attributes, providing interpretable data support for subsequent sensory evaluation and laying a quantitative foundation for expert review and Skill encapsulation.

3.4. Agent Workflow Implementation

This section focuses on the system's underlying technical implementation, covering four core dimensions of the Agent: the orchestration framework, the memory mechanism, the tool-invocation protocol, and fault tolerance and observability.

3.4.1. Workflow Orchestration Framework: LangGraph + State Machine

LangGraph is used as the core orchestration engine, defining the execution path among subsystems as a directed graph rather than a traditional linear chain of calls.

Nodes: the four subsystems (omics analysis, model invocation, report

graph.

rendering, method discovery) are encapsulated as independent, reusable graph nodes, each with a clearly defined input schema and output schema.

Edges: define the transition conditions between nodes and support conditional edges — for example, when model-evaluation confidence falls below a threshold, execution branches to an exception-handling node instead of proceeding directly to report rendering.

State management: LangGraph's built-in StateGraph maintains a global state object that carries data as it is passed and transformed between nodes, ensuring the entire process is traceable.

Rationale: compared with LangChain's Chain pattern, LangGraph offers more complete support for loops, branching, and state management, making it better suited to complex scenarios involving coordination among multiple subsystems.

3.4.2. Agent Memory Mechanism: Tiered Storage Strategy

The Agent needs to maintain context across multi-turn tasks to avoid redundant computation and information loss. A three-tier memory architecture is used:

Memory Tier	Storage Medium	Lifecycle	Purpose
Short-term memory (working area)	In-memory dictionary / Redis	Single task session	Stores intermediate variables for the current analysis task (e.g., the sample ID currently being processed, the latest model output)
Long-term memory (knowledge	Vector database (Chroma /	Persistent	Stores historical analysis cases

base)	Milvus)		and expert revision records for RAG-based retrieval augmentation, assisting decisions on new tasks
Structured memory (metadata)	PostgreSQL	Persistent	Stores task execution traces, model versions, and data lineage, for auditing and reproduction

Key mechanism: at the start of every task, the Agent retrieves similar historical cases via vector similarity search and injects the 3–5 most relevant historical execution records into the prompt as context, improving decision accuracy.

3.4.3. Tool-Invocation Protocol: MCP Standardized Interface

To resolve the protocol inconsistency that arises when the Agent calls external tools (database queries, model-training scripts, report generators), all tools are uniformly encapsulated using the MCP (Model Context Protocol) specification.

Tool registration: each tool is deployed independently as an MCP Server, declaring its input parameters, output format, and invocation constraints (such as timeout and concurrency limits) via a JSON Schema.

Dynamic discovery: on startup, the Agent pulls the list of currently available tools from a service registry (Consul/Nacos) via the MCP Client, enabling hot-swappable tools.

Invocation example: the Agent's decision outputs a structured instruction {"tool": "opls_da", "params": {...}}; the MCP gateway handles routing, authentication, and rate limiting, and returns the result to the Agent in a unified wrapper.

Rationale: the MCP protocol decouples tools from the Agent, so adding a new analysis tool requires no changes to the Agent's code — only registering a new MCP Server.

3.4.4. Fault Tolerance and Retry Mechanism

The Agent's execution chain is long, and failure at any node could interrupt the task. The system is designed with multiple layers of fault tolerance:

Node-level retry: using the tenacity library, each graph node is configured with exponential-backoff retry (initial wait 1 s, multiplier 2, maximum 10 s), up to 3 retries — effective against occasional network timeouts or model API rate limiting.

Circuit breaking and degradation: when a node fails repeatedly beyond a threshold (e.g., 5 failures within 5 minutes), a circuit breaker trips, the Agent skips that node, and returns a preset fallback response (e.g., “the analysis service is currently busy, please try again later”).

Checkpointing: using LangGraph's persistence capability, a state snapshot is saved to Redis after each key node executes. If a task is interrupted abnormally, it can resume from the most recent checkpoint, avoiding redundant computation.

3.4.5. Observability System: Full-Chain Tracing

OpenTelemetry combined with LangSmith provides visualized monitoring of the Agent's entire runtime process:

Observability Dimension	Implementation	Purpose
Trace tracking	LangSmith automatically captures every LLM call and tool execution	Debugging prompt effectiveness, locating failed nodes
Metrics collection	OpenTelemetry instrumentation feeding into	Monitoring task success rate, average duration, and token

	Prometheus/Grafana	consumption
Log aggregation	Structured JSON logs output to ELK	Post-hoc auditing and issue tracing

The complete text of the chain-of-thought for every task execution is also recorded, making it easy to later analyze whether the Agent's reasoning logic met expectations.

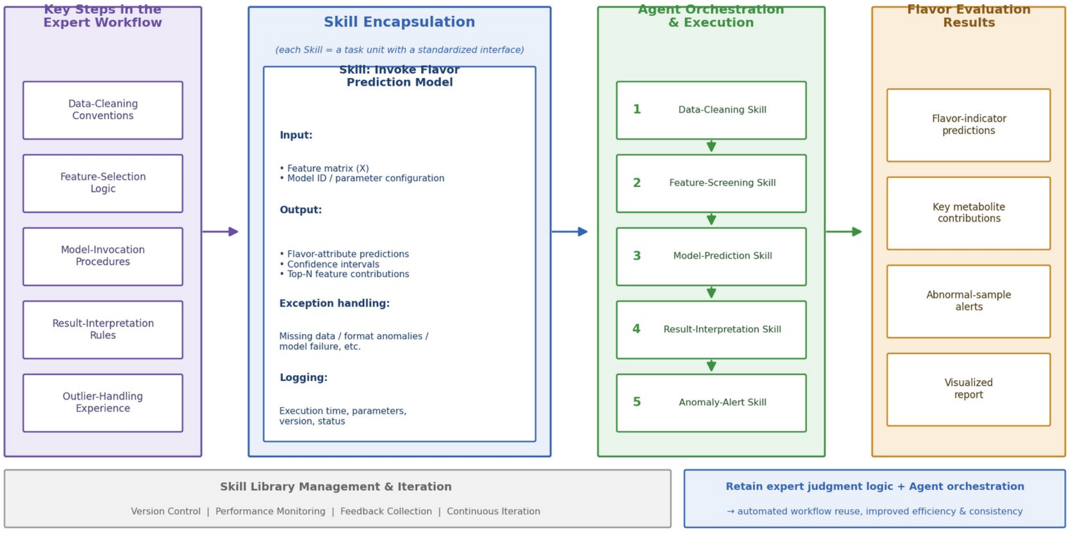
3.4.6. Deployment Architecture

Agent service: deployed as stateless containers (Kubernetes Pods) supporting horizontal scaling to handle concurrent analysis requests.

Model service: the open-source model (Qwen2.5-7B) is deployed using the vLLM framework for high-throughput inference; the closed-source model (GPT-4o) is called via API, with backup keys configured for automatic failover.

Tool service: each MCP Server is deployed independently, decoupled from the Agent service, and can be scaled individually.

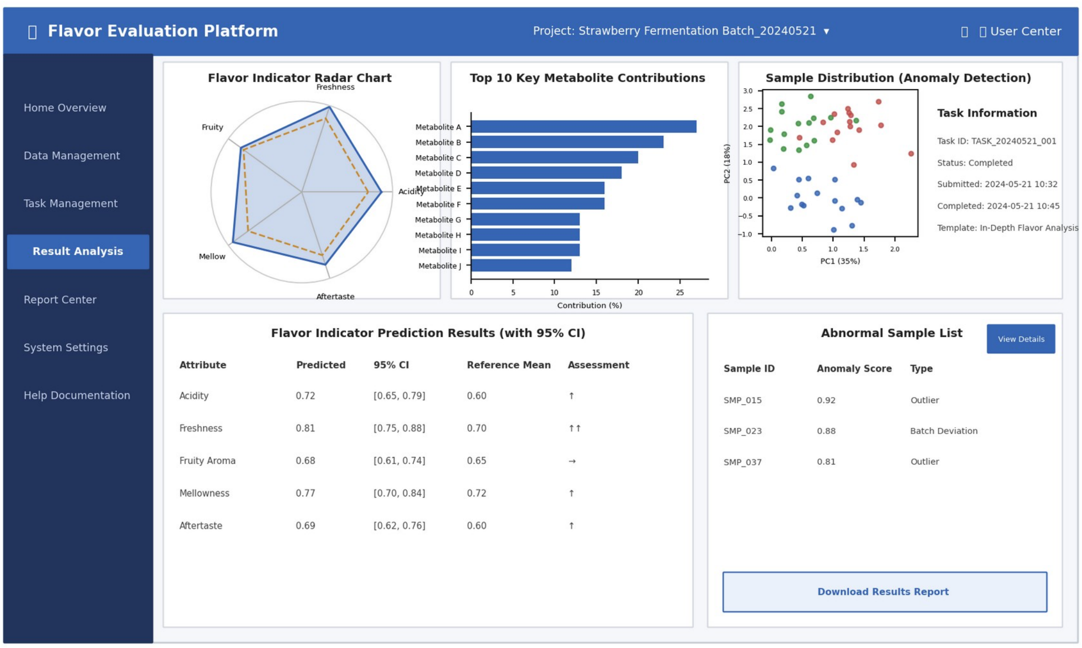
Building on the machine-learning flavor-prediction model, the system encapsulates the key steps of the aforementioned expert workflow — including data-cleaning conventions, feature-selection logic, model-invocation procedures, result-interpretation rules, and outlier-handling experience — into standardized-interface Skills that can be called by a large language model (LLM) Agent (Figure 3).



In concrete implementation, each Skill corresponds to a clearly defined task unit (such as "execute metabolite feature screening," "invoke the flavor-prediction model," "generate a flavor radar chart," "output abnormal-sample alerts," etc.), and comes with structured input/output specifications, error-handling mechanisms, and execution logging. When a new batch of upstream metabolomic data arrives, the Agent automatically orchestrates and calls the corresponding Skills in series or in parallel according to task requirements, completing the automated computation and interpretation from raw data to flavor indicators. This encapsulation approach preserves human experts' judgment logic at key decision points while leveraging the Agent's orchestration capability to achieve automated reuse of the analysis workflow, significantly improving the efficiency and consistency of flavor-report generation. At the same time, the Skill library is extensible — individual Skills can be iteratively updated based on expert feedback or new data characteristics without rebuilding the entire pipeline.

3.5. User-Friendly Interface Development

To make the system convenient for enterprise users and to provide a general-purpose platform that lowers the cost of adoption, the system is paired with an intuitive, easy-to-use visual interactive interface. The interface is built on a Web architecture; users need no programming or data-analysis background to complete the entire workflow — data upload, task submission, progress tracking, and report download — through a browser, and the interface is also adapted for mobile devices for convenient inspection by engineers. The interface is organized into the following functional modules: (1) Data Upload & Management Module — supports batch import of mass-spectrometry data files with automatic validation of data-format integrity; (2) Analysis Task Configuration Module — users can select analysis templates (e.g., "Quick Flavor Screening," "In-Depth Flavor Analysis") according to actual needs and set parameter preferences; (3) Results Display & Visualization Module — presents flavor-prediction results through linked charts, including flavor radar charts, ranked key-metabolite contributions, abnormal-sample flags, and confidence intervals; (4) Report Generation & Export Module — supports one-click generation of standardized flavor-evaluation reports (PDF/Excel format), together with a detailed comparison of raw data and prediction results. In addition, the interface includes built-in guidance and online help documentation to reduce the learning curve, and reserves a feedback channel for collecting optimization suggestions from front-line users — allowing the platform to remain general-purpose while accommodating the individual usage habits of different enterprises, truly achieving the goal of "ready to use out of the box, lightweight deployment."



4. Commercialization and Implementation

SWOT Analysis

I. Strengths

(1) Compared with traditional flavor-evaluation methods:

Traditional flavor-quality judgment relies heavily on veteran masters' experience-based assessment — “look, smell, taste, touch” — lacking an objective, quantifiable, unified standard (Sensory Evaluation Practices); at the same time, manual sensory evaluation is markedly subjective, prone to fatigue, and lagging (Sensory Evaluation of Food: Principles and Practices). Research and industry practice show that human smell and taste exhibit clear sensory adaptation and fatigue under repeated, continuous stimulation, reducing perceptual sensitivity and evaluation consistency, with judgments easily affected by emotional state, physical condition, and fatigue level (Sensory Evaluation of Food: Principles and Practices; Descriptive Sensory Analysis: Practice). For this reason, food sensory evaluation typically needs to limit the number of samples assessed at a time, using measures such as spacing between samples and palate cleansing to reduce the interference of sensory fatigue on results (Sensory Evaluation Practices).

At the same time, senior tasters, vinegar tasters, and brewing engineers in the condiment industry tend to skew older in age, talent development cycles are long, and the transmission of experience-based expertise faces real challenges (China Food Industry Yearbook). Training a qualified master capable of reliably handling core flavor-judgment tasks typically requires years of practical accumulation, at considerable time and labor cost. This evaluation system, which relies